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FINAL TERM PAPER:

**RANG-AY: An AgriHub Integrating Agricultural Development and
Flood-Responsive Community Infrastructure in Urdaneta City,
Pangasinan**

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As rural economies across the Philippines confront the combined pressures of agricultural modernization, climate variability, and uneven access to infrastructure, community-centered facilities are increasingly recognized by local governments as important assets that can strengthen livelihoods, promote cooperation, and drive development. In Urdaneta City, Pangasinan, one of the major development thrusts identified in the Comprehensive Land Use Plan is the strengthening of the city's agricultural and agro-industrial identity through the consolidation of support services and the building of disaster-resilient communities. This vision is intended to complement the city's emergence as a regional economic hub, as it would offer an integrated environment for farmer education, cooperative trading, and livelihood development. However, despite being addressed in the CLUP, the consolidation of agricultural services remains only partially realized. The dispersal of training, storage, and market facilities across multiple locations, together with the recurring impact of flooding on agricultural communities, highlights the need to strengthen Urdaneta City's agricultural identity through architecture that can function as a mediator between productivity and resilience.

In the case of Urdaneta City, the proposed RANG-AY AgriHub presents a similar opportunity to reconcile agricultural advancement with disaster preparedness, ensuring that development does not come at the expense of community safety. Inspired by the Ilocano word “*rang-ay*”, meaning growth and embodying the values of progress, prosperity, and collective advancement that reflect the deep agricultural character of Northern Luzon, the consolidation of agricultural support services, community facilities, and flood-responsive design strategies including adaptive reuse of spatial typologies, elevated structural systems, and integrated community spaces can strengthen Urdaneta City's identity as an agro-industrial center of Northern

Luzon while promoting sustainable rural development, farmer empowerment, and long-term resilience.

The Province of Pangasinan is considered one of the most productive agricultural regions in the Philippines. Like other localities in the province, Urdaneta City has a deeply agricultural past. Established as a municipality in the mid-nineteenth century and named after Father Miguel de Urdaneta, the city evolved from a small settlement along historic trade routes into a major commercial and agricultural hub in Northern Luzon. Its strategic location at the convergence of regional road networks contributed to its emergence as a center for trade, post-harvest activity, and agricultural exchange. These geographic and historical conditions shaped the active participation of Urdaneta's farming communities in the regional economy and established the city's identity as a vital node in the agricultural supply chain of Northern Luzon.

The agricultural significance of Urdaneta is most evident in its sustained role in food production, agro-industrial processing, and rural livelihood. The city hosts active farming communities whose productivity supports both local consumption and regional distribution. According to the city's Comprehensive Land Use Plan, Urdaneta functions as an important agricultural and agro-industrial center, with agriculture remaining central to its economic character and cultural identity (Urdaneta City Government, 2021). Over the years, Urdaneta transformed from a rural municipality into a highly urbanized city through rapid commercial, residential, and infrastructural development. Despite this growth, the city continues to recognize its agricultural foundations through livelihood programs, cooperative activities, and the preservation of farming traditions, reflecting the enduring importance of agriculture and community enterprise in shaping the character of Urdaneta.

Given Urdaneta City's deep agricultural identity as a productive hub of Northern Luzon, the city presents a unique opportunity to leverage community-centered architecture as a driver of rural development. Rather than allowing agricultural services to remain dispersed and underutilized, an integrated architectural approach consolidates education, livelihood, and disaster preparedness within a single environment, ensuring that productivity and resilience can coexist. This approach recognizes agricultural communities not only as beneficiaries of infrastructure but also as active participants in shaping spaces that respond to their everyday and emergency needs (Cruz & Forman, 2022). Community facilities are therefore treated not merely as containers of function but as active instruments of social and economic empowerment.

According to Till (2024), participatory architecture must accommodate the unpredictable realities of its users, meaning rural facilities must be flexible enough to respond to the cyclical and seasonal nature of farming. His work on the dependency of architecture on social conditions emphasizes that meaningful community space emerges when design acknowledges the lived rhythms and contingencies of its users. Furthermore, resilience-oriented architecture can create social, economic, and environmental benefits while strengthening a community's capacity to withstand and recover from disruption. Recent studies on community disaster resilience argue that infrastructure planning must extend beyond everyday function to anticipate the spatial and logistical demands of emergencies (Cutter et al., 2014). This perspective highlights the importance of creating environments that not only support routine agricultural activity but also sustain community safety during extreme weather events.

In the context of Urdaneta City, the proposed RANG-AY AgriHub reflects the principles of community-centered and resilience-oriented architecture. The facility seeks to consolidate agricultural support services while introducing flexible learning spaces, cooperative trading areas, flood-adaptive structural systems, and emergency-ready community environments. These interventions align with established strategies that emphasize the integration of livelihood, participation, and disaster preparedness. By strengthening the accessibility of agricultural services and reinforcing the city's capacity to respond to flooding, the AgriHub can reinforce Urdaneta City's agricultural identity while contributing to sustainable and resilient rural development.

The proposed RANG-AY AgriHub is guided by the Ilocano word “*rang-ay*”, signifying progress, prosperity, and collective advancement. As a cultural value deeply rooted in the agricultural communities of Northern Luzon, rang-ay reflects the belief that progress is achieved not in isolation but through shared effort and mutual upliftment. In the context of agricultural development, rang-ay represents the act of bringing together the city's farming communities in order to strengthen their present productivity and guide their long-term resilience. Rather than viewing agricultural infrastructure as a purely functional asset, the concept encourages the reinterpretation and reintegration of community, livelihood, and security into a single architectural environment.

One of the most significant opportunities within the proposed AgriHub is the consolidation of dispersed agricultural services into a single centralized facility. According to the Comprehensive Land Use Plan of Urdaneta City, agricultural development and agro-industrial growth are central to the city's long-term vision (Urdaneta City Government, 2021). As services

remain scattered across multiple locations, the AgriHub presents an opportunity to integrate training, post-harvest processing, storage, and market access within a unified environment. Rather than allowing these functions to remain fragmented, the facility can be established as a focal point of the city's agricultural enterprise. Through centralized planning, the AgriHub can serve as a primary hub that strengthens farmer education, cooperative trading, and livelihood development, while improving the accessibility and efficiency of agricultural support.

Architecture plays a key role in shaping community identity and collective welfare. Beyond productivity, the built environment carries social narratives, shared experiences, and the everyday rhythms of farming life. Community-centered design, therefore, is not only about housing functions but also about communicating the meaning of a place through its spaces, gathering areas, and civic interactions. As an agro-industrial center, Urdaneta possesses a distinct agricultural character that can be translated into the design of its community spaces. Instead of relying solely on segregated facilities, social and livelihood functions can be expressed through multi-purpose halls, cooperative offices, learning centers, and assembly areas that allow community life to unfold within the same environment as everyday agricultural activity. By integrating these social and economic layers into the architectural fabric, the AgriHub becomes more than a service facility; it becomes a living environment that strengthens community bonds, supports civic participation, and keeps Urdaneta's agricultural identity visible and relevant.

Establishing flood-adaptive systems that allow the facility to remain functional during and after extreme weather events is necessary for ensuring both livelihood continuity and community safety. According to climate adaptation frameworks, rural infrastructure must anticipate recurring

hazards and maintain essential operations despite disruption (Asian Development Bank [ADB], 2021; Intergovernmental Panel on Climate Change [IPCC], 2023). These measures may include elevated building platforms, reinforced concrete structural systems, flood-resistant materials, adaptable interior spaces, and clearly defined circulation routes for emergency response. These guidelines emphasize that protecting not only daily operations but also the capacity to shift into emergency function is essential in maintaining the resilience and social value of agricultural facilities. This approach reinforces the role of flood-responsive design in sustaining livelihood continuity and preventing disruption within vulnerable farming settings.

Sustainable design further strengthens the environmental performance and long-term viability of the facility. Rather than treating sustainability as an additive feature, the AgriHub integrates passive cooling, natural ventilation, daylighting, rainwater harvesting, solar energy systems, bioswales, and permeable pavements as core components of its architectural logic. These strategies support environmental responsibility while reducing long-term operational costs and promoting resilient rural development. By embedding sustainability within the everyday functioning of the facility, the AgriHub demonstrates how agricultural infrastructure can reinforce the city's identity as a progressive and responsible agro-industrial center of Northern Luzon.

Ultimately, the AgriHub exemplifies how architecture can become an instrument of collective progress, transforming the principles of *rang-ay* into a lasting framework for resilience, empowerment, and sustainable rural development. The concept symbolizes the act of collective progress of bringing together the farming communities whose labor sustains the city. Through the consolidation of agricultural support services, the adaptive integration of community and

livelihood spaces, the adoption of flood-responsive structural systems, and the application of sustainable design measures, the AgriHub can better support, protect, and empower the agricultural communities of Urdaneta.

By transforming agricultural infrastructure from a passive logistical asset into an active part of daily community life, these interventions enable farmers, residents, and local enterprises to interact more deeply with the spaces that sustain their livelihoods. More than consolidating services, these strategies create a distinct and recognizable environment that distinguishes Urdaneta from other rapidly urbanizing cities. As a result, the AgriHub can serve not only as a center for agricultural productivity and disaster preparedness but also as a lasting symbol of Urdaneta City's agricultural significance and its enduring role in the region's journey toward sustainable rural development. Rang-ay demonstrates how collective progress can become a catalyst for community empowerment, ensuring that the agricultural foundations of Urdaneta remain visible, secure, and deeply embedded within the city's future.

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